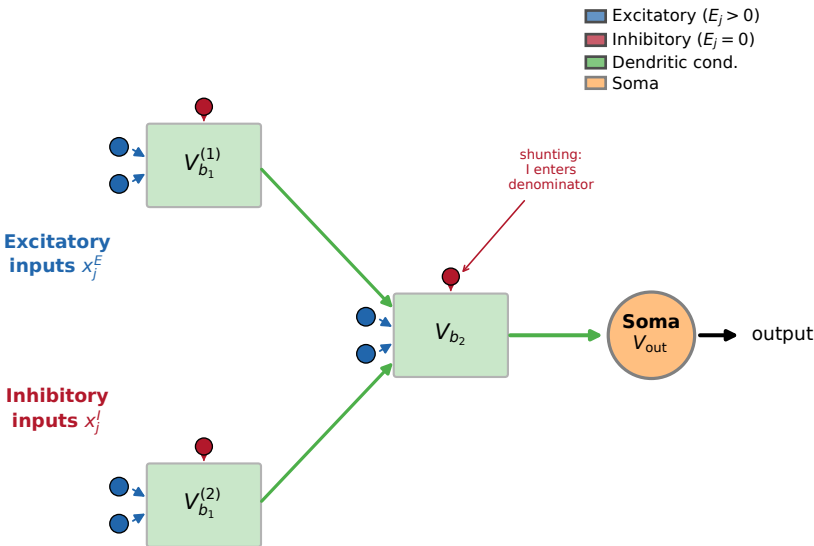
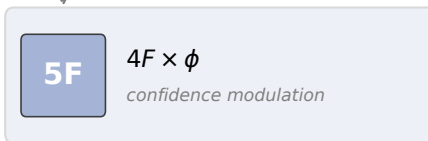
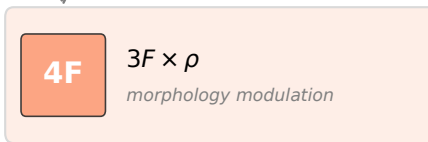
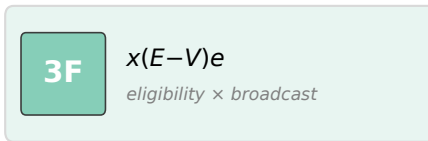


Compartmental dendritic neuron



Local learning rule hierarchy

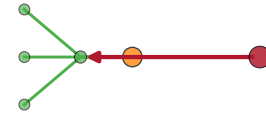
base LocalCA family



Broadcast modes

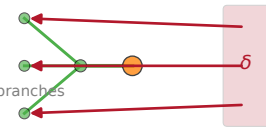
scalar

one shared field



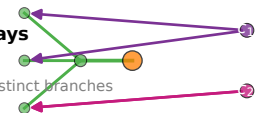
per-soma

vector from soma to all branches



structured pathways

separate channels for distinct branches



Exact factorization summary

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = [x_i R_n^{\text{tot}} (E_i - V_n)] \cdot \left[\frac{\partial L}{\partial V_n} \right]$$

exact gradient = local eligibility × compartment error

Weighted cosine

1.000001

± 0.000001 across conditions

Max scale mismatch

2.6e − 08

all theory-diagnostic conditions

Only approximation in LocalCA: $e_n \approx \partial L / \partial V_n$